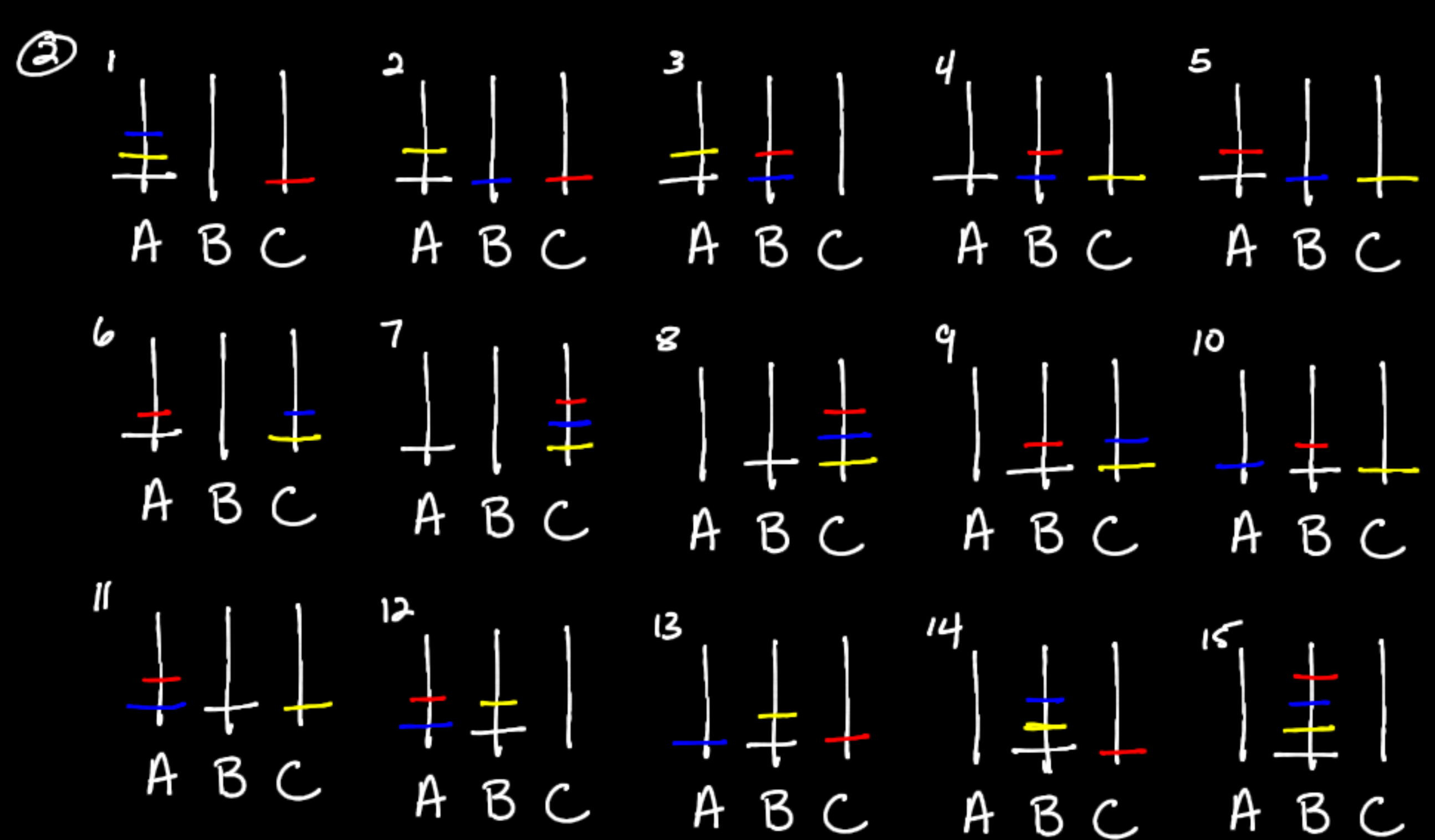
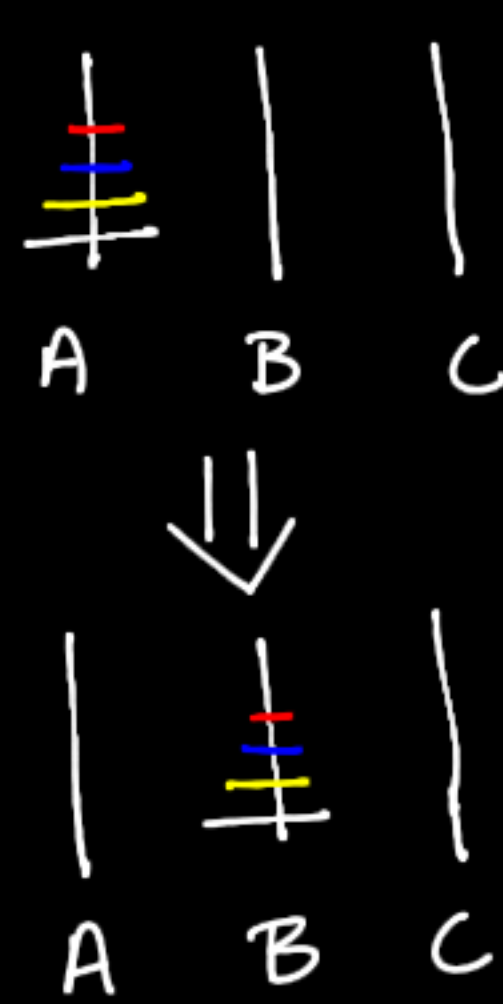
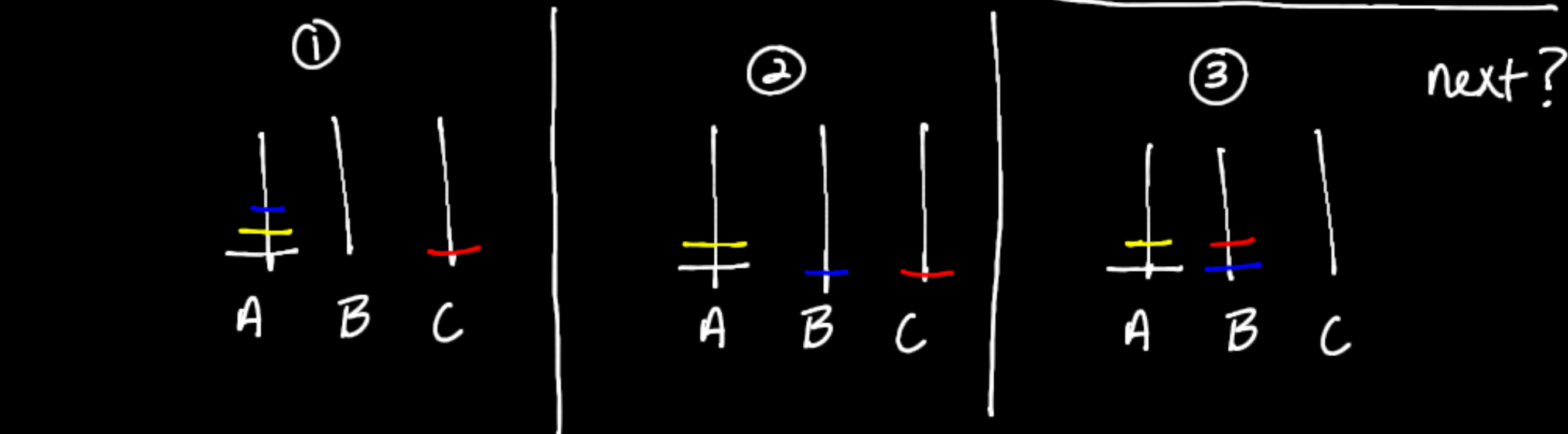


- ① Towers of Hanoi  
 n disks of n different sizes  
 Stacked by size on a pole  
 3 poles - source, destination, spare  
 move only one disk at a time  
 Cannot stack a larger disk on a smaller  
 goal is to move all disks to the destination  
 minimize the number of moves



Note: "source," "destination," and "spare" change for each move



- ③ Think about the steps (also see the book)
1. Ignore bottom disk, solve  $n-1$  disks w/ spare B and destination C
  2. move the disk from A to B
  3. move disks from C w/ spare A and destination B

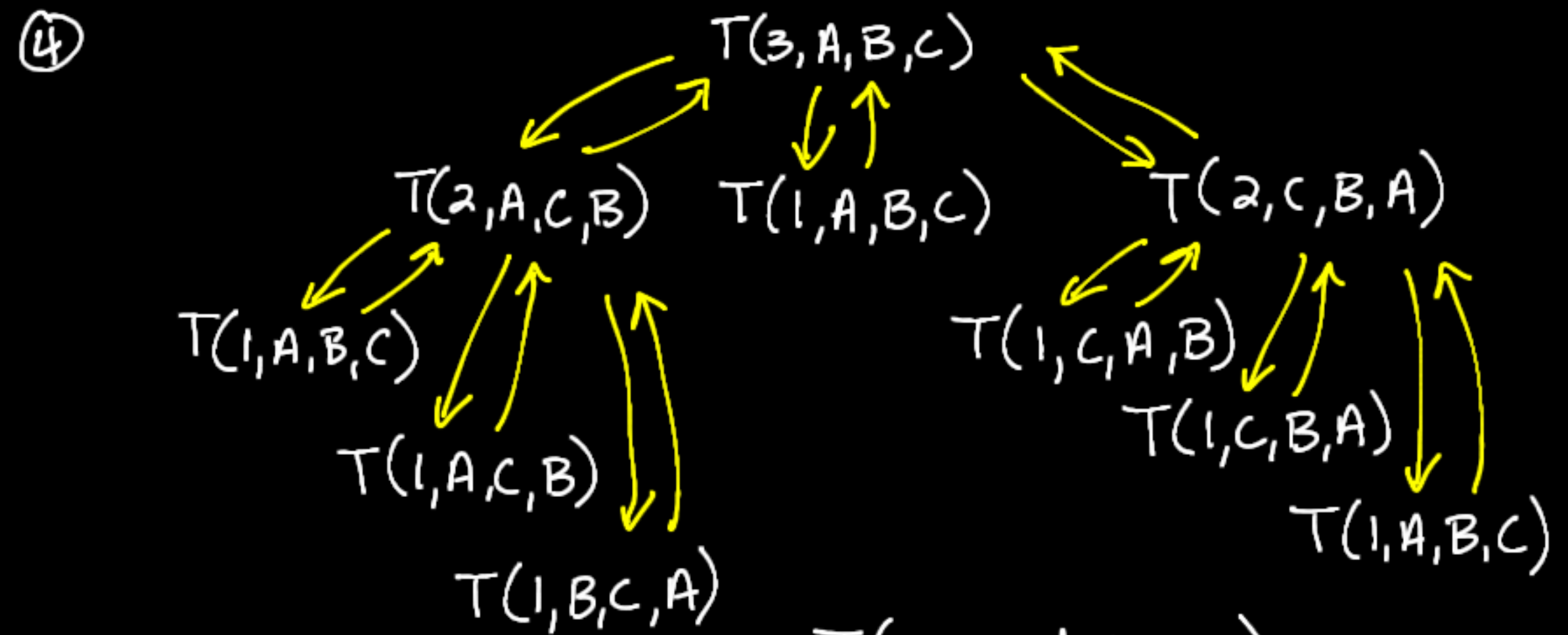
1. Towers( $n-1, A, C, B$ )  
 all but bottom of A → source destination spare



2. Towers( $1, A, B, C$ ) ← spare is bottom disk  
 bottom disk → src dest spare not used



3. Towers( $n-1, C, B, A$ )  
 all but bottom of C → src dest spare



T( $n, \text{src}, \text{dest}, \text{spr}$ )  
 if ( $n==1$ )  
 print(move top disk from src to dest)  
 else  
 T( $n-1, \text{src}, \text{spr}, \text{dest}$ )  
 T( $1, \text{src}, \text{dest}, \text{spr}$ )  
 T( $n-1, \text{spr}, \text{dest}, \text{src}$ )